EXAM-ORIENTED STUDY GUIDE

Broad View of Operating Systems

MS-DOS • UNIX • MS-WINDOWS

MS-DOS **UNIX** **MS-WINDOWS**

A complete, structured revision guide covering definitions, history, architecture, features, file systems, commands, and comparisons of the three major operating systems — designed for competitive examinations and teacher recruitment examinations.

DEFINITIONS **ARCHITECTURE DIAGRAMS** **COMPARISON TABLES** **QUICK REVISION NOTES** **PRACTICE MCQs**

Compiled Study Material | Computer Science / Computer Awareness | Operating Systems Module

INTRODUCTION

Broad View of Operating Systems

An **Operating System (OS)** is system software that acts as an interface between the computer user and computer hardware. It manages hardware resources (CPU, memory, storage, I/O devices) and provides services so that application programs can run efficiently. Historically, the evolution of operating systems can be broadly viewed through three landmark systems — **MS-DOS**, **UNIX**, and **MS-WINDOWS** — each representing a distinct stage in OS design philosophy: from simple single-user command-line systems, to powerful multi-user multitasking systems, to modern graphical multitasking systems.

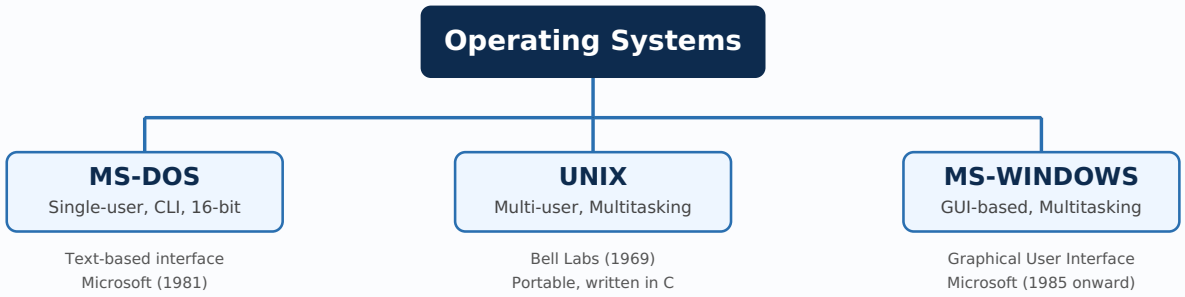


Fig 0.1 — Broad classification: MS-DOS, UNIX, MS-WINDOWS

DEFINITION

An Operating System is a program that acts as an intermediary between the user of a computer and the computer hardware, controlling and coordinating the use of hardware among various application programs.

Why These Three Systems Matter

System	Era / Type	Significance
MS-DOS	1981 — Single-user, CLI	One of the earliest widely-used PC operating systems; foundation of early personal computing; purely command-driven.
UNIX	1969 — Multi-user, Multitasking	Pioneered portability (written in C), hierarchical file systems, and multi-user/multitasking concepts that influenced almost all modern OS design including Linux and macOS.
MS-WINDOWS	1985 onward — GUI-based, Multitasking	Brought the Graphical User Interface (GUI) to the mass market; became the dominant desktop OS worldwide.

EXAM POINT

Operating systems are broadly classified by: (a) number of users — Single-user vs Multi-user; (b) number of tasks — Single-tasking vs Multitasking; (c) interface type — CLI (Command Line Interface) vs GUI (Graphical User Interface). MS-DOS is single-user/single-tasking/CLI; UNIX is multi-user/multitasking/CLI (with optional GUI); MS-WINDOWS is single/multi-user, multitasking, and GUI-based.

CHAPTER 1

MS-DOS (Microsoft Disk Operating System)

1.1 Introduction and Definition

DEFINITION

MS-DOS (Microsoft Disk Operating System) is a single-user, single-tasking, command-line-based operating system developed by Microsoft for IBM-compatible personal computers. It was the primary operating system for PCs throughout the 1980s and early 1990s before the rise of GUI-based Windows.

MS-DOS operates entirely through typed text commands at a prompt (commonly shown as **C:\>**) rather than through icons or windows. The user must know exact command names and syntax to perform tasks such as copying files, deleting files, or running programs.

1.2 History and Evolution

- Developed by **Microsoft**, MS-DOS was based on **86-DOS** (also called QDOS — "Quick and Dirty Operating System"), purchased from Seattle Computer Products.
- First released on **August 12, 1981** as **PC-DOS/MS-DOS 1.0** with IBM's first personal computer (IBM PC).
- It went through multiple versions — MS-DOS 2.0 (1983, introduced hierarchical directories), MS-DOS 3.0, 4.0, 5.0, up to **MS-DOS 6.22** (final standalone retail version, 1994).
- Later versions of MS-DOS were bundled "underneath" early Windows versions (Windows 3.x, 95, 98) until Windows became independent of DOS from Windows NT/2000/XP onward.

1.3 Architecture of MS-DOS

MS-DOS is the textbook example of a **Simple Structure** — one of the basic ways an operating system can be organized. The diagram below shows MS-DOS split into four levels, but this is *not* the same as a true Layered Structure (covered for UNIX in Chapter 2). The two ideas are easy to mix up, so the difference is explained right after the diagram.

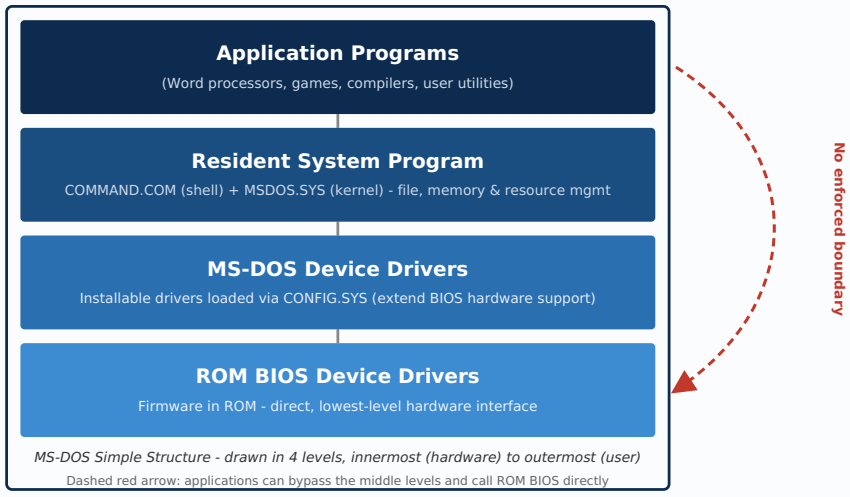


Fig 1.1 — Simple structure of MS-DOS (drawn in 4 levels)

The Four Levels, Bottom to Top

Level	What It Does
1. ROM BIOS Device Drivers	Firmware burnt into ROM by the hardware manufacturer; the most basic, lowest-level routines for talking to the keyboard, screen, disk, etc.
2. MS-DOS Device Drivers	Extra drivers loaded from CONFIG.SYS at boot to support hardware the ROM BIOS alone cannot handle.
3. Resident System Program	The DOS kernel and shell (MSDOS.SYS + COMMAND.COM) that stay resident in memory and provide file, memory, and command-processing services.
4. Application Programs	User-run software — word processors, games, compilers — sitting at the top, the level the end user directly interacts with.

EXAM POINT — SIMPLE STRUCTURE VS. LAYERED STRUCTURE

Both MS-DOS and UNIX are drawn as stacked levels, so exams often test whether you can tell them apart by their *rule*, not their picture:

- **True Layered Structure (UNIX):** a strict rule is enforced — a level may call *only* the level directly below it. The user can never reach hardware except by passing through every layer in between.
- **Simple Structure (MS-DOS):** there is no such enforcement. An application program is free to skip levels and call the ROM BIOS directly, bypassing the kernel and device drivers entirely. Because the boundaries between levels are not strictly policed, a fault in one application can bring down the whole system.

This is precisely why MS-DOS is labelled a **Simple Structure** even though its diagram has four levels, while UNIX earns the name **Layered Structure**.

EXAM POINT — LAYER ORDER

A very common MCQ asks you to arrange the MS-DOS levels from innermost to outermost. Correct order: **ROM BIOS Device Drivers → MS-DOS Device Drivers → Resident System Program → Application Programs**.

Core System Files

File	Function
IO.SYS	Handles low-level input/output operations; interfaces directly with BIOS and hardware drivers.
MSDOS.SYS	The kernel of MS-DOS; manages files, executes programs, and handles error processing and resource allocation.
COMMAND.COM	The command interpreter / shell; reads and executes user-typed commands (internal commands run directly; external commands load as separate .COM/.EXE files).
BIOS (ROM)	Basic Input Output System, stored in firmware (ROM); the lowest-level software that initializes hardware and provides MS-DOS access to it.

MEMORY TRICK

Remember the three core files using **"I-M-C"**: **I**O.SYS (Input/Output) → **M**SDOS.SYS (Main kernel) → **C**OMMAND.COM (Command interpreter / shell, the part the user sees). For the 4 architecture levels, remember **"R-D-R-A"**: **R**OM BIOS → **D**OS Device Drivers → **R**esident System Program → **A**pplication Programs.

1.4 Features and Characteristics

- **Single-user:** Only one user can operate the system at a time.
- **Single-tasking:** Only one program/process can run at a time (no true multitasking).
- **Command Line Interface (CLI):** No graphical icons; user types commands.
- **16-bit operating system** designed for Intel 8086/8088 processors.
- **Real-mode operation:** Does not support protected/virtual memory mode natively.
- **Limited memory addressing:** Maximum of 1 MB addressable RAM, with only about 640 KB available to user programs (the rest reserved for system use) — this 640 KB limit is famously called the **"DOS memory barrier."**

- Supports **batch files** (.BAT) for automating sequences of commands.
- No built-in networking or security/multi-user support.
- Simple, lightweight, and requires very little hardware/storage to run.

1.5 File System and File Naming

MS-DOS uses the **FAT (File Allocation Table)** file system — specifically **FAT12/FAT16** — to organize and track data stored on disk.

File Naming Convention — "8.3 Format"

> **KEY RULE**
> MS-DOS file names follow the **8.3 naming convention**: a maximum of **8 characters** for the file name, followed by a period (.), followed by a maximum of **3 characters** for the file extension. Example: **READ_ME.TXT**, **AUTOEXEC.BAT**, **COMMAND.COM**.

- Spaces and most special characters are not allowed in file names.
- File names are not case-sensitive.
- Directory structure is hierarchical (tree-like), introduced from MS-DOS 2.0 onward.

Common File Extensions in MS-DOS

Extension	Meaning
.COM	Command / executable file (small machine-code programs)
.EXE	Executable program file
.BAT	Batch file (sequence of DOS commands)
.SYS	System file (device drivers, core OS files)
.TXT	Plain text file

1.6 Important MS-DOS Commands

DOS commands are classified into two types: **Internal Commands** (built into COMMAND.COM, loaded into memory at boot) and **External Commands** (separate executable files stored on disk).

Internal Commands

Command	Function
DIR	Displays list of files and directories
COPY	Copies one or more files to another location
DEL / ERASE	Deletes one or more files
REN	Renames a file
MD / MKDIR	Creates (makes) a new directory
RD / RMDIR	Removes an (empty) directory
CD / CHDIR	Changes current directory
CLS	Clears the screen
TYPE	Displays contents of a text file
VER	Displays the DOS version number
DATE / TIME	Displays or sets system date/time

External Commands

Command	Function
FORMAT	Formats a disk to prepare it for use
CHKDSK	Checks disk for errors and displays status report
XCOPY	Copies files and directory trees (more powerful than COPY)
DISKCOPY	Copies the entire contents of one floppy disk to another
LABEL	Creates, changes, or deletes a disk's volume label
ATTRIB	Displays or changes file attributes (read-only, hidden, etc.)
TREE	Displays directory structure graphically

EXAM POINT

A simple test often used in exams: Internal commands run **without needing a separate file on disk** (they're already loaded in memory with COMMAND.COM), whereas external commands need their own **.COM/.EXE file present on disk** to execute.

1.7 Memory Management in MS-DOS

- MS-DOS uses a simple **single contiguous memory allocation** scheme.
- Total addressable memory in real mode = **1 MB**, divided as:
- **Conventional Memory (0–640 KB):** Available for DOS, drivers, and application programs.
- **Upper Memory Area / UMA (640 KB–1 MB):** Reserved for BIOS, video memory, and hardware.

Later add-ons like **EMM386** and HIMEM.SYS allowed access to Expanded Memory (EMS) and Extended Memory (XMS) beyond the 1 MB limit, but the core 640 KB barrier remained a famous limitation of native DOS.

1.8 Advantages and Disadvantages of MS-DOS

Advantages

- Simple, lightweight; requires very little memory/storage
- Fast execution since there is no GUI overhead
- Full direct control over hardware via commands
- Stable for running single dedicated tasks
- Low system requirements; runs on very old hardware

Disadvantages

- No multitasking — only one program at a time
- No multi-user support
- Difficult for beginners (must memorize commands)
- No built-in networking or internet support
- 8.3 file naming is highly restrictive
- No graphical interface; no mouse support natively
- Limited to 1 MB addressable memory (real mode)

QUICK REVISION NOTES — MS-DOS

Identity & History

- Full Form: **Microsoft Disk Operating System**
- Type: **Single-user, Single-tasking**, Command Line Interface (CLI), **16-bit**
- Released: **August 12, 1981** as PC-DOS/MS-DOS 1.0, derived from 86-DOS / QDOS
- Last standalone version: **MS-DOS 6.22** (1994); later versions ran "underneath" Windows 3.x/95/98

Architecture

- OS structure type: **Simple Structure** (no enforced layering; apps can call BIOS directly)
- 4 levels (innermost → outermost): **ROM BIOS Device Drivers → MS-DOS Device Drivers → Resident System Program → Application Programs**
- Core files: **IO.SYS** (I/O), **MSDOS.SYS** (kernel), **COMMAND.COM** (shell)

File System & Commands

- File system: **FAT (FAT12/FAT16)**; naming rule: **8.3 format** (8-char name + 3-char extension)
- Commands: **Internal** (in COMMAND.COM, e.g. DIR, COPY, CD) vs **External** (separate .COM/.EXE file, e.g. FORMAT, CHKDSK, XCOPY)

Memory & Trade-offs

- Memory: **640 KB** usable (Conventional) out of **1 MB** addressable (real mode) — the "DOS memory barrier"
- Biggest strength: lightweight and fast; biggest weakness: no multitasking/multi-user/networking support
- Successor: replaced in mainstream use by GUI-based MS-Windows

CHAPTER 2

UNIX Operating System

2.1 Introduction and Definition

DEFINITION

UNIX is a powerful, multi-user, multitasking operating system originally developed at **Bell Labs** (AT&T — American Telephone and Telegraph Company) that introduced concepts such as a hierarchical file system, pipes, and portability through implementation in a high-level language (C), making it one of the most influential operating systems in computing history.

UNIX was designed to allow **many users** to run **many programs simultaneously**, sharing CPU time and resources efficiently through time-sharing. Its design philosophy — "everything is a file" and "write programs that do one thing and do it well" — heavily influenced later systems including Linux, BSD, Solaris, and macOS (Darwin is UNIX-based).

2.2 History and Evolution

- Developed in **1969** at **Bell Laboratories (AT&T)** by **Ken Thompson** and **Dennis Ritchie**.
- Originally written in assembly language; rewritten in **1973** in the **C programming language** (also developed by Dennis Ritchie), which made UNIX highly **portable** across different hardware platforms.
- The name "UNIX" is a pun on "Multics," an earlier, more complex multi-user OS project that UNIX's developers had worked on before creating a simpler alternative.
- Licensed to universities and companies, leading to many variants: BSD (Berkeley Software Distribution), System V (AT&T), Solaris (Sun Microsystems), AIX (IBM), HP-UX (Hewlett-Packard), and others.
- Inspired the creation of **Linux** (by Linus Torvalds, 1991) as a free, open-source UNIX-like OS.

2.3 Architecture of UNIX

UNIX follows a **layered structure** — the standard reference model for UNIX system architecture. Each layer uses only the services of the layer directly below it: hardware forms the base, the kernel sits directly above hardware, system utilities/shell sit above the kernel, and application programs/users form the outermost layer.

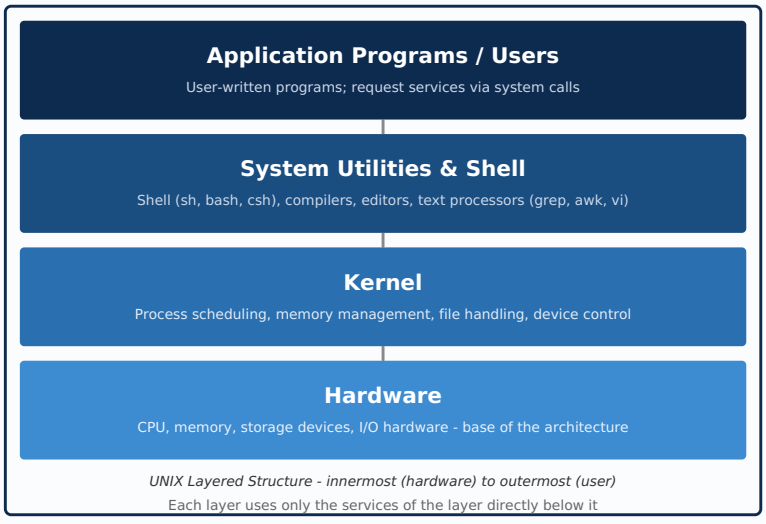


Fig 2.1 — Layered structure of UNIX (Hardware → Kernel → Shell/Utilities → Applications)

Major Architectural Components (bottom to top, as in Fig 2.1)

Component	Description
Hardware	CPU, memory, storage devices, and I/O hardware; the base layer on which the entire UNIX architecture rests.
Kernel	The core of the OS; directly interacts with hardware. Manages process scheduling, memory management, device/I-O management, and the file system. It is loaded into memory at boot and remains resident.
Shell / System Utilities	The shell is the command interpreter that sits between the user and the kernel, translating typed commands into kernel system calls. Common shells: **sh (Bourne Shell), csh (C Shell), ksh (Korn Shell), bash (Bourne Again Shell)**. System utilities (compilers, editors, text processors like grep, awk, sed, vi) sit in this same layer.
Application Programs / Users	The outermost layer; user-written programs request services from the layers below via system calls.

MEMORY TRICK

"H-K-S-A" — **H**ardware (innermost/base) → **K**ernel (talks to hardware) → **S**hell/Utilities (interpreter, talks to user) → **A**pplication Programs (outermost layer).

EXAM POINT

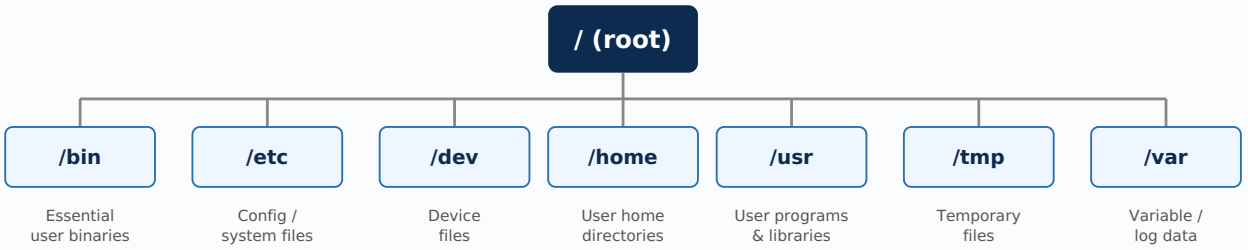
UNIX is the textbook example of a true **Layered Structure**, in contrast to MS-DOS's **Simple Structure** (see Section 1.3 for the full Simple-vs-Layered distinction). The short version: in UNIX, a layer may call *only* the layer directly below it — there is no shortcut to hardware.

2.4 Features and Characteristics

- **Multi-user:** Multiple users can log in and use the system simultaneously (via terminals or networks).
- **Multitasking:** Multiple processes can run concurrently using CPU time-sharing.
- **Portability:** Written largely in C, allowing UNIX to run on diverse hardware with minimal changes.
- **"Everything is a file":** Devices, directories, and processes are all represented and accessed as files (the file system itself is detailed in Section 2.5).
- **Pipes and Redirection:** Output of one command can be used as input to another using the pipe symbol (**|**), redirection using **>** and **<**.
- **Built-in security:** File permissions (read/write/execute) for owner, group, and others; user authentication via login/password.
- **Built-in networking** support (TCP/IP) — historically one of the first OSes with strong networking capability.
- **Shell scripting:** Supports writing scripts to automate tasks.
- Available with both **CLI** and (in many distributions) optional GUI environments.

2.5 UNIX File System

UNIX uses a single **hierarchical (tree-structured) file system** with a single root directory denoted by a forward slash **"/"**. All storage devices are "mounted" onto this single tree rather than being given separate drive letters (unlike MS-DOS/Windows, which use C:, D:, etc.).



UNIX uses a single hierarchical (tree-structured) file system rooted at "/"

Every file/device is represented as a node in this tree — "Everything is a file"

Fig 2.2 — UNIX hierarchical file system structure

Directory	Purpose
/	Root directory — top of the file system hierarchy
/bin	Essential binary (executable) programs needed for the system to operate
/etc	System configuration files
/dev	Device files representing hardware devices
/home	Home directories for individual users
/usr	User programs, libraries, and documentation
/tmp	Temporary files (often cleared on reboot)
/var	Variable data such as logs and spool files

File Permission System

Every UNIX file has three sets of permissions — for the **owner**, the **group**, and **others** — each consisting of **read (r), write (w), execute (x)** rights, often shown as a 10-character string, e.g. **-rwxr-xr--**.

> **EXAM POINT**
> UNIX permission notation: **r=4, w=2, x=1** — added together to give the familiar numeric (octal) permission codes, e.g. **chmod 755** means owner=rwx(7), group=r-x(5), others=r-x(5). This numeric permission system is a common exam question.

2.6 Process Management

- Every running program in UNIX is a **process**, identified by a unique **Process ID (PID)**.
- The first process started at boot is called **"init"** (PID 1), which is the parent of all other processes.
- UNIX supports both **foreground processes** (interactive, requires user wait) and **background processes** (run independently, denoted with **&**).
- Processes can create child processes using the **fork()** system call; **exec()** replaces a process image; **wait()** makes a parent wait for a child to finish.

2.7 Important UNIX Commands

Command	Function

ls	Lists files and directories
cd	Changes current directory
pwd	Prints (displays) current working directory
cp	Copies files
mv	Moves or renames files
rm	Removes (deletes) files
mkdir / rmdir	Creates / removes a directory
cat	Displays or concatenates file contents
grep	Searches for a text pattern within files
chmod	Changes file permissions
chown	Changes file ownership
ps	Displays currently running processes
kill	Terminates a running process
man	Displays the manual/help page for a command

2.8 UNIX Variants / Flavours

Variant	Developed By
System V	AT&T (American Telephone and Telegraph Company)
BSD (Berkeley Software Distribution)	University of California, Berkeley
Solaris	Sun Microsystems (later Oracle)
AIX	IBM
HP-UX	Hewlett-Packard
Linux (UNIX-like, not UNIX itself)	Linus Torvalds (open-source community)

2.9 Advantages and Disadvantages of UNIX

Advantages

- True multi-user and multitasking support
- Highly stable and reliable; rarely crashes
- Strong built-in security (permissions, user authentication)
- Portable across hardware platforms (written in C)
- Powerful built-in networking capabilities
- Flexible shell scripting for automation

Disadvantages

- Steep learning curve; not beginner-friendly
- Command syntax can be cryptic/terse
- Different variants/flavours can cause compatibility issues
- Historically limited commercial GUI support compared to Windows
- Can require more skilled administrators

QUICK REVISION NOTES — UNIX

Identity & History

- Type: **Multi-user, Multitasking** OS; CLI by default (optional GUI in many distributions)
- Developed: **1969**, Bell Labs (AT&T), by **Ken Thompson & Dennis Ritchie**
- Rewritten in **C language** in 1973 → gave it portability across hardware
- Name is a pun on **"Multics"**, an earlier, more complex OS project

Architecture

- OS structure type: **Layered Structure** (strict layering, unlike MS-DOS's Simple Structure)
- 4 layers (innermost → outermost): **Hardware → Kernel → System Utilities & Shell → Application Programs/Users**
- Common shells: **sh, csh, ksh, bash**

File System & Permissions

- File system: single hierarchical tree rooted at **"/"**; devices are mounted onto it (no drive letters)
- Philosophy: **"Everything is a file"**
- Permissions: **read(4) / write(2) / execute(1)** for owner/group/others, e.g. chmod 755

Processes & Variants

- Every process has a unique **PID**; **"init"** is the first process (PID 1) and parent of all others
- Process calls: **fork()** creates a child, **exec()** replaces a process image, **wait()** pauses the parent
- Major variants: **BSD, System V, Solaris, AIX, HP-UX**; inspired the free/open-source **Linux**

Trade-offs

- Biggest strength: stability, security, and portability; biggest weakness: steep learning curve for beginners
- Influenced: Linux, BSD, Solaris, macOS (Darwin)

CHAPTER 3

MS-WINDOWS

3.1 Introduction and Definition

DEFINITION

MS-Windows (Microsoft Windows) is a family of **Graphical User Interface (GUI)-based** operating systems developed by Microsoft, designed to make computers easier to use through visual elements like windows, icons, menus, and a pointer (mouse), instead of requiring users to type text commands.

Unlike MS-DOS, Windows allows users to interact with the computer visually — by clicking icons, dragging windows, and using drop-down menus — and supports **multitasking**, allowing several applications to run and be visible simultaneously, each within its own "window" (hence the name).

3.2 History and Versions

MS-Windows evolved through two parallel lines before merging into one: a **consumer line** (Windows 1.0 through 98/ME) that ran on top of MS-DOS, and a separate **business line** built on the **NT kernel** (Windows NT, 2000) that was independent of DOS. The two lines merged at **Windows XP**.

Version	Year	Key Highlight
Windows 1.0	1985	First release; a graphical shell layered on top of MS-DOS
Windows 2.0	1987	Introduced overlapping windows
Windows 3.0 / 3.1	1990 / 1992	First major commercial success; still ran on top of DOS
Windows 95	1995	Introduced the Start button, taskbar, and Plug and Play — major leap in usability
Windows 98 / ME	1998 / 2000	Continued the consumer (9x) line; still DOS-based
Windows NT / 2000	1993 / 2000	Separate business-oriented line on the new NT kernel — not dependent on DOS
Windows XP	2001	Merged the consumer and business lines onto the NT kernel; extremely stable and long-lived
Windows Vista / 7	2007 / 2009	Refined GUI, security, and performance
Windows 8 / 8.1	2012 / 2013	Touch-focused redesign with the Start screen
Windows 10	2015	Unified platform across devices; Cortana, Edge browser
Windows 11	2021	Redesigned interface with a centered Start menu

EXAM POINT

Windows 1.0 (1985) was Microsoft's first GUI-based operating environment, but it ran on top of MS-DOS. True independence from DOS came with the **NT kernel** family (Windows NT, 2000, XP, and later). This DOS → NT-kernel transition is a frequently tested concept.

3.3 Architecture of MS-Windows

Modern Windows (NT-based) architecture is divided into two main modes: **User Mode** and **Kernel Mode**, separated for stability and security.

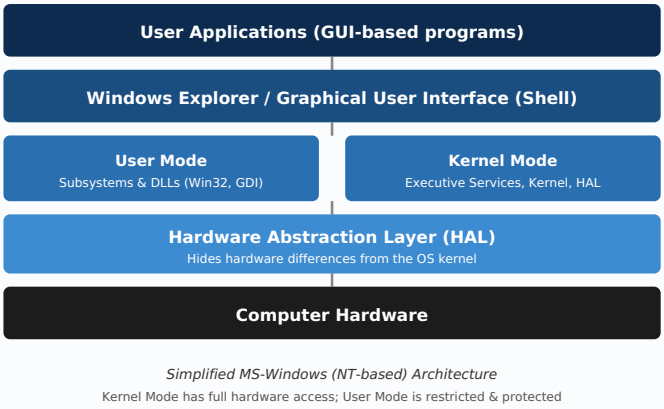


Fig 3.1 — Simplified MS-Windows (NT-based) architecture

Component	Description
User Mode	Restricted execution environment where applications and subsystems run; cannot directly access hardware. Includes Win32 subsystem, GDI (Graphics Device Interface).
Kernel Mode	Privileged mode with full hardware access; contains the Windows Executive, the Kernel itself, and device drivers.
HAL (Hardware Abstraction Layer)	A software layer that hides hardware-specific details from the rest of the OS, improving portability across different hardware.
GUI Shell (Windows Explorer)	Provides the visual desktop environment — icons, taskbar, file explorer windows.

3.4 Features and Characteristics

- **Graphical User Interface (GUI):** icons, windows, menus, pointers — easy for beginners.
- **Multitasking:** multiple applications can run simultaneously, each in its own window.
- **Plug and Play (PnP):** automatic detection and configuration of hardware devices.
- **Multi-user support** (in modern versions) with separate user accounts and permissions.
- **Built-in networking** and Internet connectivity support.
- **Wide software/hardware compatibility** — by far the most widely supported desktop OS for third-party software and drivers.
- **Backward compatibility** with many older applications.
- Supports **plug-in peripherals**, multimedia, and modern security features (Windows Defender, BitLocker, User Account Control).
- Storage uses **drive letters** (C:, D:, etc.) — covered in detail in Section 3.5.

3.5 File System

File System	Notes
FAT16 / FAT32	Older file systems, inherited from DOS-era Windows; FAT32 supports larger drives than FAT16 but has a 4 GB max single file size limit.
NTFS (New Technology File System)	Modern default file system for Windows; supports file/folder permissions, encryption, compression, larger volumes/files, and journaling for reliability.

Windows organizes storage using **drive letters** (e.g., C:\ for the primary system drive), with each drive/partition having its own independent directory tree — a key structural difference from UNIX's single unified tree.

3.6 Advantages and Disadvantages of MS-Windows

Advantages

- User-friendly GUI; easy for beginners
- Extremely wide software and hardware support
- Strong multitasking and multimedia capability
- Plug and Play simplifies hardware setup
- Large user community and support resources

Disadvantages

- Higher hardware/resource requirements than DOS
- More vulnerable to viruses/malware due to widespread use
- Licensing costs (proprietary, not free/open-source)
- Can be resource-heavy compared to lightweight CLI systems

QUICK REVISION NOTES — MS-WINDOWS

Identity & History

- Type: **GUI-based**, Multitasking; supports single and multiple user accounts (modern versions)
- First release: **Windows 1.0 in 1985** (ran atop MS-DOS)
- Major usability leap: **Windows 95** (Start button, taskbar, Plug and Play)
- Independent kernel line: **Windows NT** (not dependent on DOS) → merged with consumer line at **Windows XP**

Architecture

- **User Mode** (restricted, runs apps/subsystems) vs **Kernel Mode** (privileged, full hardware access)
- **HAL (Hardware Abstraction Layer)** hides hardware differences from the kernel, improving portability
- GUI shell: **Windows Explorer** provides the visual desktop environment

File System & Storage

- Modern default file system: **NTFS** (permissions, encryption, journaling); older: **FAT16/FAT32**
- Storage organization: **drive letters** (C:, D:...) — each partition has its own tree, unlike UNIX's single unified tree

Trade-offs

- Key feature: **Plug and Play** automatic hardware detection; widest software/hardware compatibility of the three systems
- Biggest strength: ease of use for beginners; biggest weakness: heavier resource needs and higher malware exposure

CHAPTER 4

Comparative Study

4.1 MS-DOS vs UNIX vs MS-WINDOWS — Master Comparison Table

This master table consolidates the most exam-relevant distinguishing points across all three operating systems for quick side-by-side revision.

Parameter	MS-DOS	UNIX	MS-WINDOWS
Developer	Microsoft	Bell Labs (AT&T) — Ken Thompson & Dennis Ritchie	Microsoft
Year of Origin	1981 (Aug 12)	1969	1985
OS Structure Type	Simple Structure (no strict layering; apps can call BIOS directly)	Layered Structure (Hardware → Kernel → Shell/Utilities → Applications)	User Mode / Kernel Mode split (NT architecture), connected via HAL
Interface	Command Line Interface (CLI)	CLI (with optional GUI in some distributions)	Graphical User Interface (GUI)
Number of Users	Single-user	Multi-user	Single/Multi-user (modern versions support multiple accounts)
Multitasking	No (single-tasking)	Yes	Yes
File System	FAT (FAT12/FAT16)	Hierarchical tree, single root "/"	FAT16/FAT32 (older), NTFS (modern)
File Naming	8.3 format (8 char name + 3 char extension)	Long names allowed; case-sensitive	Long file names allowed; not case-sensitive
Storage Organization	Drive letters (A:, C:)	Single unified tree; devices "mounted" onto it	Drive letters (C:, D:, etc.)
Security	Minimal/none	Strong (permissions, authentication)	Moderate to strong (modern versions: accounts, UAC, encryption)
Portability	Limited (tied closely to x86 PC hardware)	High (written in C)	Primarily x86/x64 PC hardware (also ARM in newer versions)
Networking	Not built-in	Strong built-in networking (TCP/IP)	Strong built-in networking and Internet support
Bit Architecture	16-bit	Varies by implementation (historically 16/32-bit; modern UNIX variants 64-bit)	Originally 16-bit; modern versions 32-bit/64-bit
Ease of Use	Difficult for beginners (must memorize commands)	Difficult for beginners; steep learning curve	Easy for beginners (icon/menu-driven)
Resource Requirement	Very low	Moderate	High (compared to DOS/lightweight UNIX)

MEMORY TRICK

"D-U-W" timeline order: **U**NIX (1969) came first → **D**OS (1981) next → **W**indows (1985) last. Despite UNIX being the oldest, it remained CLI-based and multi-user, while DOS (newer than UNIX) was simpler/single-user, and Windows (newest) added the GUI.

CHAPTER 5

Final Exam Capsule

5.1 One-Line Facts to Remember

Fact	Answer
Full form of MS-DOS	Microsoft Disk Operating System
Year MS-DOS was released	1981 (August 12, 1981)
MS-DOS originally based on	86-DOS / QDOS
Three core files of MS-DOS	IO.SYS, MSDOS.SYS, COMMAND.COM
File system used by MS-DOS	FAT (FAT12/FAT16)
File naming rule of MS-DOS	8.3 format
Maximum usable conventional memory in DOS	640 KB
OS structure type of MS-DOS	Simple Structure
MS-DOS architecture layers (innermost → outermost)	ROM BIOS → DOS Device Drivers → Resident System Program → Application Programs
UNIX developed at	Bell Labs (AT&T)
UNIX developed by	Ken Thompson & Dennis Ritchie
Year UNIX was developed	1969
Language UNIX was rewritten in (1973)	C language
OS structure type of UNIX	Layered Structure
UNIX architecture layers (innermost → outermost)	Hardware → Kernel → Shell/Utilities → Application Programs
UNIX root directory symbol	/ (forward slash)
UNIX design philosophy	"Everything is a file"
First process in UNIX (PID 1)	init
First version of MS-Windows released	Windows 1.0, 1985
Windows version that introduced Start button/taskbar	Windows 95
Modern default Windows file system	NTFS
Windows hardware-hiding layer	HAL (Hardware Abstraction Layer)
Windows kernel family independent of DOS	Windows NT

5.2 Practice MCQs with Answers

Q1. MS-DOS is classified as a:

(a) Multi-user, Multitasking OS (b) Single-user, Single-tasking OS (c) Multi-user, Single-tasking OS (d) Network OS

Answer: (b) Single-user, Single-tasking OS

Q2. Which file in MS-DOS acts as the command interpreter (shell)?

(a) IO.SYS (b) MSDOS.SYS (c) COMMAND.COM (d) BIOS.SYS

Answer: (c) COMMAND.COM

Q3. Arrange the MS-DOS architecture layers from innermost to outermost:

(a) Application Programs → Resident System Program → MS-DOS Device Drivers → ROM BIOS (b) ROM BIOS → MS-DOS Device Drivers → Resident System Program → Application Programs (c) MS-DOS Device Drivers → ROM BIOS → Application Programs → Resident System Program (d) Resident System Program → Application Programs → ROM BIOS → MS-DOS Device Drivers

Answer: (b) ROM BIOS → MS-DOS Device Drivers → Resident System Program → Application Programs

Q4. MS-DOS is classified under which type of OS structure?

(a) Layered Structure (b) Microkernel Structure (c) Simple Structure (d) Modular Structure

Answer: (c) Simple Structure

Q5. The MS-DOS file naming convention is commonly known as:

(a) 4.4 format (b) 16.32 format (c) 8.3 format (d) 12.16 format

Answer: (c) 8.3 format

Q6. UNIX was originally developed at:

(a) IBM (b) Bell Labs (AT&T) (c) Microsoft (d) Xerox PARC

Answer: (b) Bell Labs (AT&T)

Q7. Who are credited as the original developers of UNIX?

(a) Bill Gates & Paul Allen (b) Ken Thompson & Dennis Ritchie (c) Linus Torvalds & Richard Stallman (d) Steve Jobs & Steve Wozniak

Answer: (b) Ken Thompson & Dennis Ritchie

Q8. UNIX gained portability mainly because it was rewritten in:

(a) Assembly language (b) FORTRAN (c) C language (d) Java

Answer: (c) C language

Q9. UNIX is the textbook example of which type of OS structure?

(a) Simple Structure (b) Layered Structure (c) Virtual Machine Structure (d) Exokernel Structure

Answer: (b) Layered Structure

Q10. In the UNIX layered structure, which layer forms the base (innermost layer)?

(a) Kernel (b) Shell (c) Hardware (d) Application Programs

Answer: (c) Hardware

Q11. In UNIX, the root directory is represented by:

(a) C:\ (b) / (c) root: (d) \root

Answer: (b) /

Q12. Which OS introduced the Start button and taskbar for the first time?

(a) Windows 3.1 (b) Windows 95 (c) Windows XP (d) Windows 1.0

Answer: (b) Windows 95

Q13. The modern default file system used by MS-Windows is:

(a) FAT12 (b) FAT16 (c) NTFS (d) EXT4

Answer: (c) NTFS

Q14. Which layer in Windows architecture hides hardware differences from the OS?

(a) Kernel Mode (b) User Mode (c) HAL (d) GDI

Answer: (c) HAL (Hardware Abstraction Layer)

Q15. Which of the following operating systems uses drive letters (C:, D:) instead of a single unified directory tree?

(a) UNIX only (b) MS-DOS only (c) MS-DOS and MS-Windows (d) UNIX and MS-Windows

Answer: (c) MS-DOS and MS-Windows

Q16. The maximum conventional memory accessible to programs in standard MS-DOS (real mode) is:

(a) 256 KB (b) 512 KB (c) 640 KB (d) 1 MB

Answer: (c) 640 KB

Q17. Which command in MS-DOS is used to format a disk?

(a) CHKDSK (b) FORMAT (c) XCOPY (d) LABEL

Answer: (b) FORMAT

Q18. Among the three systems, which was developed first chronologically?

(a) MS-DOS (b) UNIX (c) MS-Windows (d) All in the same year

Answer: (b) UNIX (1969)

Q19. Which UNIX command displays the contents of the current working directory's path?

(a) ls (b) cd (c) pwd (d) cat

Answer: (c) pwd

End of Study Guide — Broad View of Operating Systems: MS-DOS, UNIX, MS-WINDOWS.